IonHack 2026

Team QByte

CounterDiabatics

The Problem: MaxCut as Ising Minimisation

NP-hard combinatorial optimisation → native Ising Hamiltonian

MaxCut Objective

Given G = (V, E, w):

Maximise $\Sigma_{(i,j)\in E}\ w_{ij}\,(1 - z_i z_j) / 2$

Ising Hamiltonian H_p:

$H_p = -\Sigma_{i<j} J_{ij} Z_i Z_j$

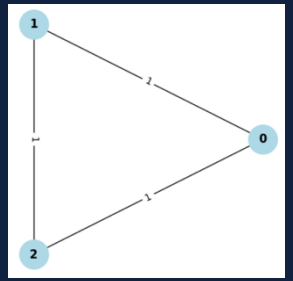
Adiabatic interpolation:
$H(\lambda) = (1-\lambda)H_0 + \lambda H_p$

Why Classical Methods Struggle

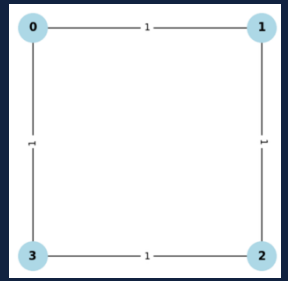
2^n exponential state space
Best classical approx ratio: 0.878 (Goemans-Williamson)

NP-hard in general; exact → exponential time

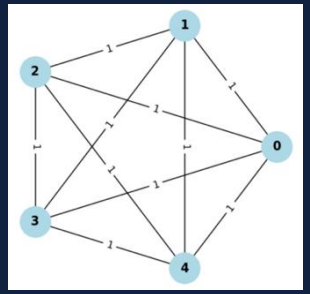
Test Graphs



Triangle, n=3



Square, n=4



Star, n=5

Methodology: End-to-End Pipeline

01

Graph → Ising Encoding

Weighted adjacency J to ZZ couplings. Brute-force classical baseline for ground truth.

02

Adiabatic Schedule

λ linearly ramped 0→1 in M=4 Trotter steps. $H(\lambda) = (1-\lambda)H_o + \lambda H_p$.

03

1st-order AGP Derivation

Introduction counterdiabatic term into Hamiltonian

04

DAQC Circuit Build

MS-gate H_p block + local H_o block + RXX CD block per step. Qiskit parameterised circuit.

05

Noisy Simulation

Depolarising on MS (p=1e-4). Density-matrix AerSimulator. 2000 shots.

06

Result Analysis

Histogram comparison DAQC vs DAQC+CD

Step 1: Problem Hamiltonian — Ising Encoding

MaxCut objective → ZZ spin interaction → native trapped-ion form

Problem Hamiltonian

$$H_p = \sum_{i<j} J_{ij}\, \sigma_i^x\, \sigma_j^x$$

where J_{ij} = edge weight (coupling strength)
σ^z = Pauli-Z operator on qubit i or j

From handwritten notes (Image 2):

Classical: $H_p = \Sigma\, J_{ij}\, \sigma_i\, \sigma_j$ (Ising spins ±1)

Quantum: $\sigma_i\, \sigma_j \rightarrow \sigma_i x\, \sigma_j x$ (XX Pauli strings)

MS gate natively implements $\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y$
→ rotate to ZZ basis via R_z gates

Mixing (Driver) Hamiltonian

$$H_0 = \sum_i \sigma_i^z$$

Transverse-field term — prepares uniform superposition, governs quantum fluctuations

Full interpolated Hamiltonian:

$$H(t) = \lambda(t) \sum_{i<j} J_{ij} \sigma_i^x \sigma_j^x + (1 - \lambda(t)) \sum_i \sigma_i^z$$

$\lambda(t)$: annealing schedule 0 → 1 over time T

$\lambda(0) = 0$ [pure mixing] $\lambda(T) = 1$ [pure problem]

Step 2: Adiabatic Schedule & Counter-Diabatic Problem

Why finite-T annealing fails — and the AGP correction

Linear λ-schedule

$$\lambda(t_k) = k \cdot \Delta t / T$$

k = 0, 1, ..., M (M Trotter steps)

$$\lambda(0) = 0, \quad \lambda(T) = 1$$

Δt = T/M (time step per Trotter layer)

The Adiabatic Bottleneck

Adiabatic theorem: stay in ground state iff T→∞
Finite T on NISQ: diabatic transitions at gap Δ(λ)
P(excitation) ~ exp(−Δ·T) → error at small T
Trotter error compounds: finite Δt introduces artefacts

Counter-Diabatic (CD) Correction

First-Oder Adiabatic Gauge Potential (AGP):

$$2\dot{\lambda}(t)\alpha_1(t)\sum_{i<j} J_{ij}\left(\sigma_j^x\sigma_i^y + \sigma_i^x\sigma_j^y\right)$$

1st-order variational truncation (Image 1 / 3):

$$\alpha_1(\lambda) = -\tfrac{1}{4}\,\Sigma J_{ij}^2 / D(\lambda)$$

Denominator D(λ):

$$D = (1-\lambda)^2 \cdot 4\Sigma J_{ij}^2 + \lambda^2 \cdot (\Sigma J_{ij}^4 + 6 \cdot \Sigma_triples)$$

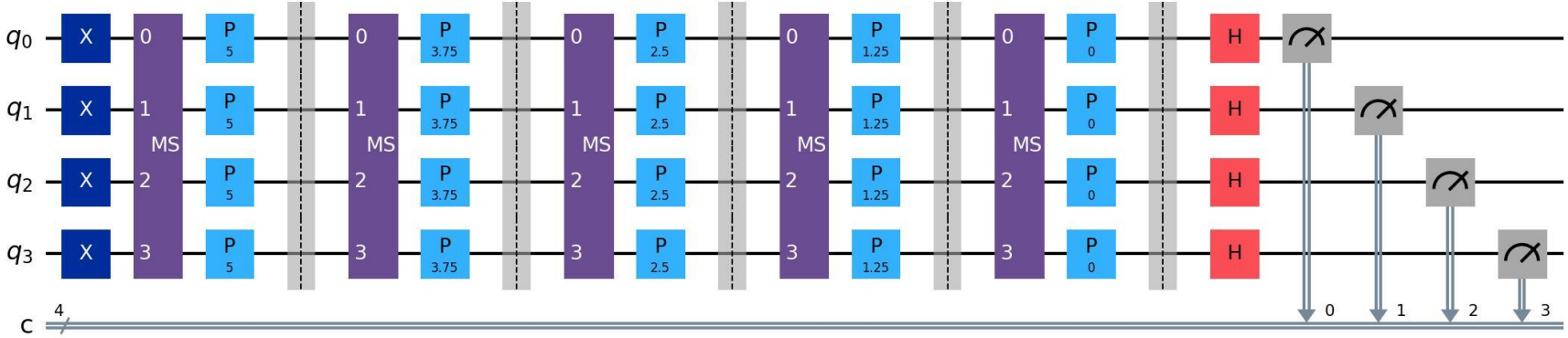
Triples term:

$$\Sigma_triples = \Sigma(J_{ij}^2 J_{ik}^2 + J_{ij}^2 J_{jk}^2 + J_{ik}^2 J_{jk}^2)$$

for all i<j<k

Step 3: Digitized quantum annealing circuit

$$\hat{U}(0,T) \approx \prod_{k=0}^{M} \prod_{n=0}^{N} \left[R_n^z\left(2(1-\lambda(t_k))\Delta t\right)\right] \mathrm{GMS}\left(\{2\lambda(t_k)\Delta t J_{ij}\}_{i<j}\right)$$

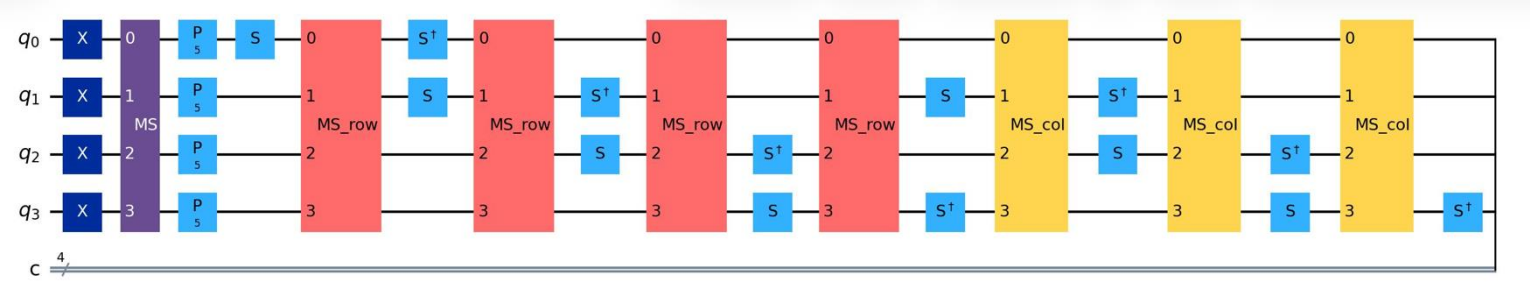


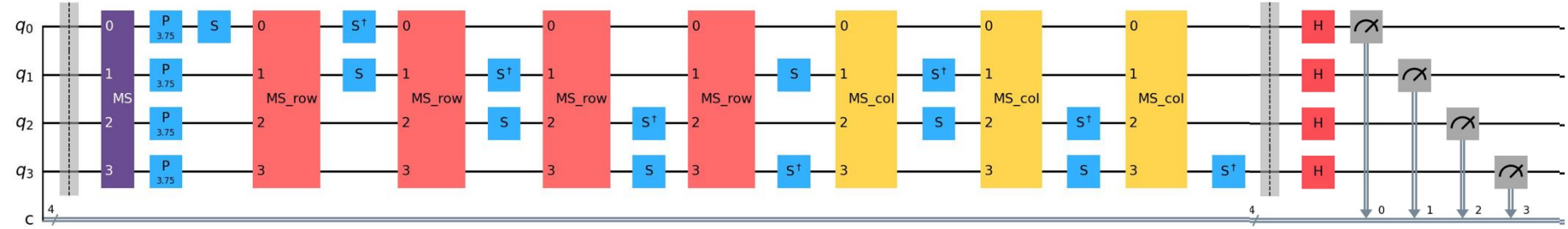
Trotter step

Step 4: Counterdiabatic contribution

$$\exp\left(-i\sum_{i<j} J'_{ij}\left(\sigma_j^x\sigma_i^y+\sigma_i^x\sigma_j^y\right)/2\right)=\left[\prod_{n=1}^{N} R_n^z\left(\frac{\pi}{2}\right)\mathrm{GMS}\left(J'_{nn+1},...,J'_{nN}\right)R_n^z\left(-\frac{\pi}{2}\right)\right]$$

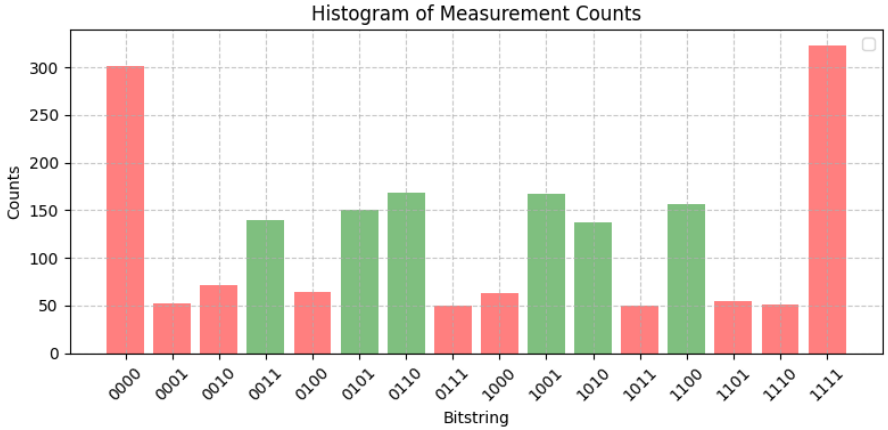
$$\times\left[\prod_{j=2}^{N} R_j^z\left(\frac{\pi}{2}\right)\mathrm{GMS}\left(J'_{1j},...,J'_{j-1j}\right)R_j^z\left(-\frac{\pi}{2}\right)\right]$$



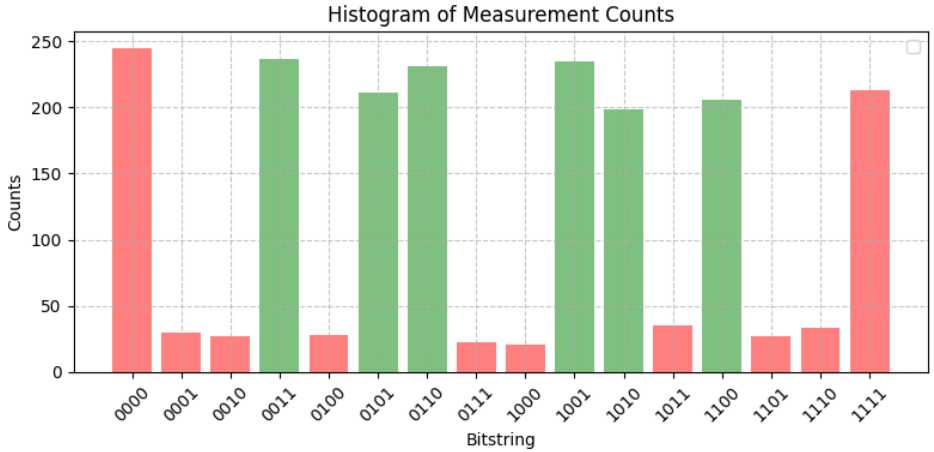


Probability distribution landscape

Square graph (n=4) with all-to-all connections, dt=1.0



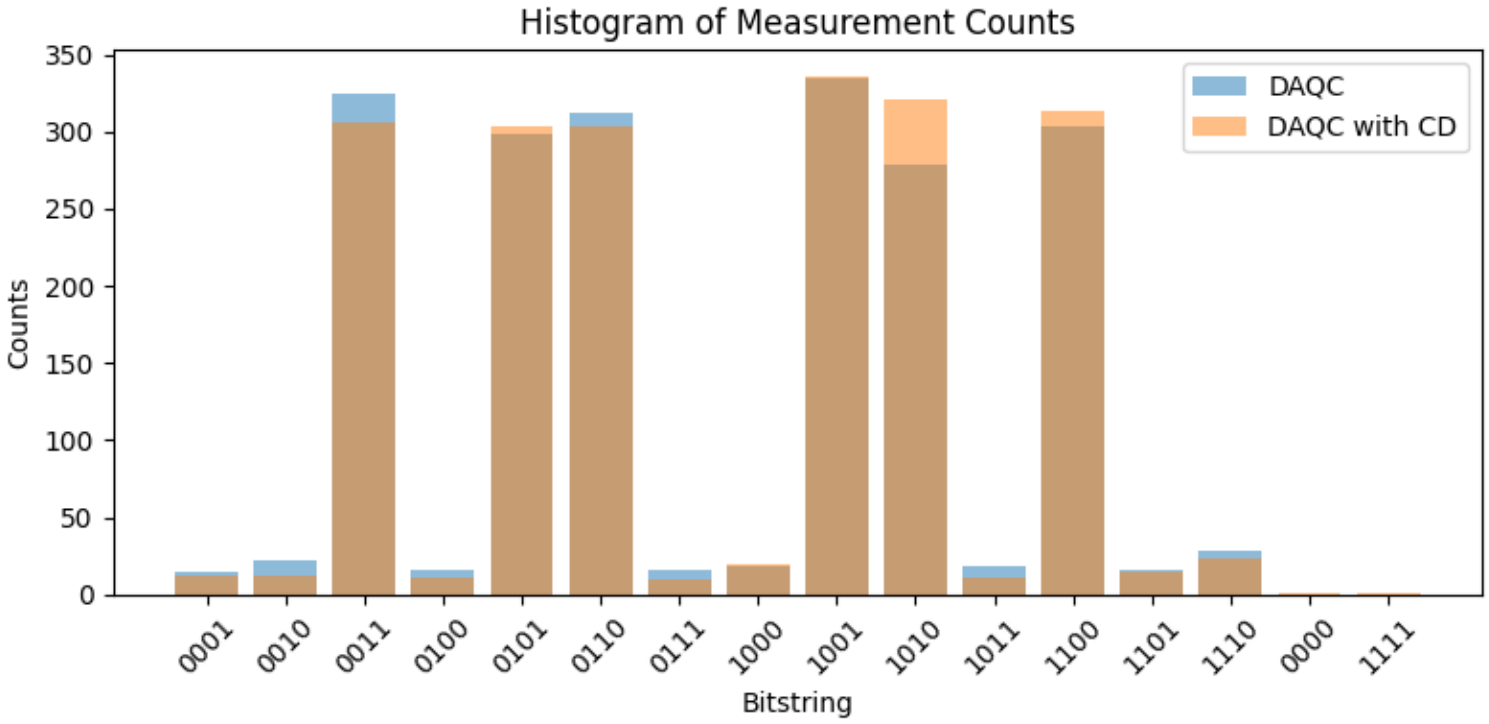
Digitized quantum annealing



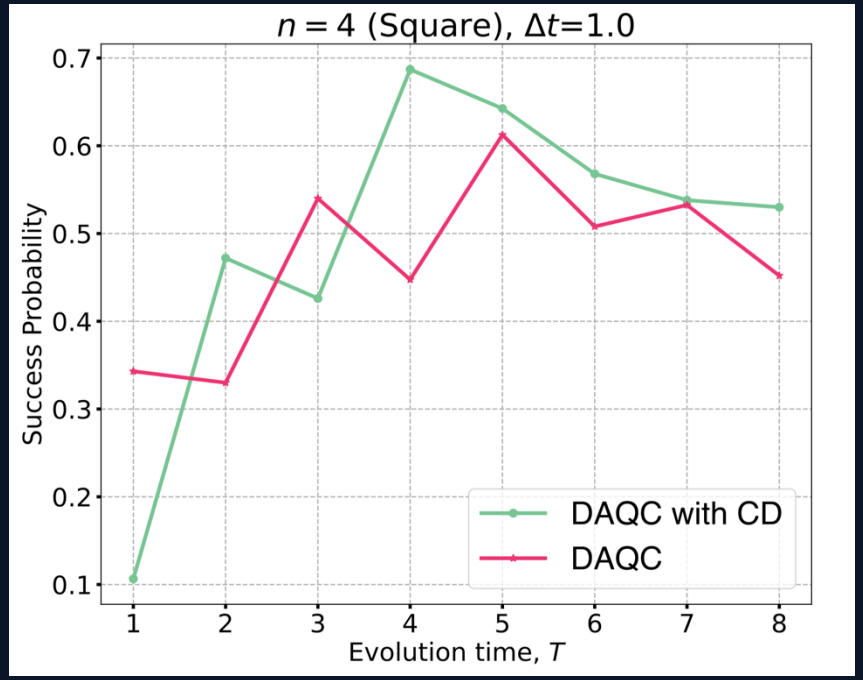
Counterdiabatic contribution

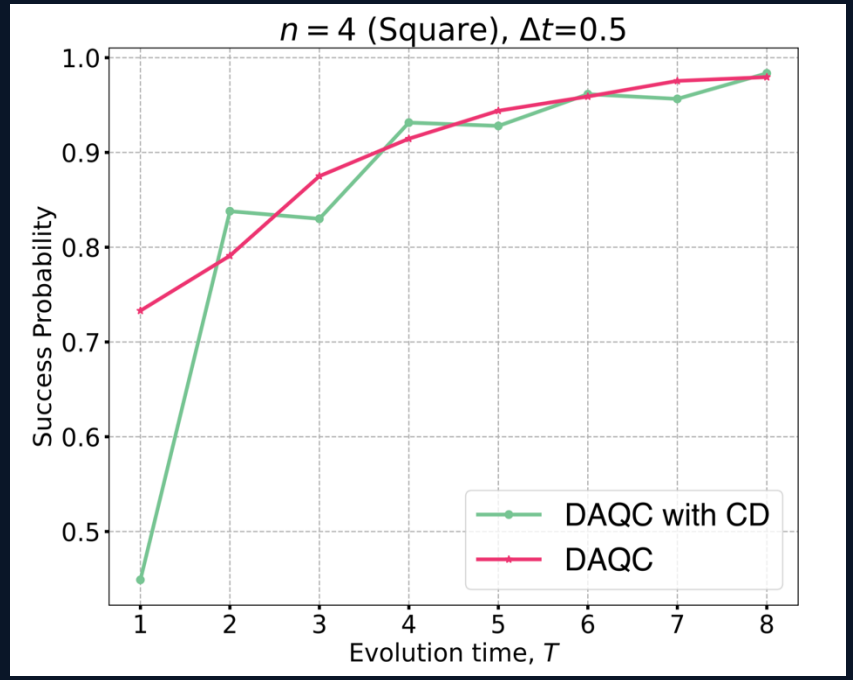
Probability distribution landscape

Square graph (n=4) with all-to-all connections, dt=0.25



CD Adavantage





Conclusions & Outlook

CD-DAQC: practical quantum advantage pathway on trapped-ion NISQ devices

Key Findings

- CD correction measurably increases optimal bitstring probability at constant circuit depth
- $\alpha_1(\lambda)$ schedule provides analytically tractable, hardware-efficient AGP approximation
- MS-gate DAQC is naturally suited to trapped-ion connectivity
- Noise resilience: p_MS = 1e-4 does not wash out CD advantage

Future Directions

- 2nd-order AGP truncation for smaller spectral gaps
- Nearest neighbours connections
- Scale to n > 10 qubits leveraging native IonQ all-to-all connectivity
- Benchmark against QAOA at equivalent gate depth on real hardware